Program 10: Creating a Simple WebApp with the help of Tomcat Server

Step 1: Run the command in order to create an project with the name "**MyMavenWebApp01**"

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp01 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false

```
sumanth@sumanth-VirtualBox:~$ cd devops
sumanth@sumanth-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp01 -DarchetypeArtifactId=maven-ar
chetype-webapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/sumanth/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenWebApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/sumanth/devops/MyMavenWebApp01
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.615 s
[INFO] Finished at: 2026-04-15T23:50:40+05:30
[INFO] ------------------------------------------------------------------------
```

Step 2: Update the pom.xml with the relevant dependencies and the relevant plugins

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>

  <artifactId>MymavenWebApp01</artifactId>

  <packaging>war</packaging>

  <version>1.0-SNAPSHOT</version>

  <name>MymavenWebApp01 Maven Webapp</name>

  <url>http://maven.apache.org</url>

 <dependencies>
```

```
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
     <!-- Servlet API for Java EE -->
        <dependency>
            <groupId>javax.servlet</groupId>
            <artifactId>javax.servlet-api</artifactId>
            <version>4.0.1</version>
            <scope>provided</scope>
        </dependency>
  </dependencies>
  <build>
  <plugins>
  <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-war-plugin</artifactId>
    <version>3.4.0</version> <!-- Update to latest stable version -->
</plugin>
</plugins>
    <finalName>MymavenWebApp01</finalName>
  </build>
</project>
```

Step 3: Run **mvn clean install** to compile and generate the **War** file

```
sumanth@sumanth-VirtualBox:~/Devops/MymavenWebApp01$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:MymavenWebApp01 >---------------------
[INFO] Building MymavenWebApp01 Maven Webapp 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MymavenWebApp01 ---
[INFO] Deleting /home/sumanth/Devops/MymavenWebApp01/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MymavenWebApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MymavenWebApp01 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MymavenWebApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/sumanth/Devops/MymavenWebApp01/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MymavenWebApp01 ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MymavenWebApp01 ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MymavenWebApp01 ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MymavenWebApp01] in [/home/sumanth/Devops/MymavenWebApp01/target/MymavenWebApp01]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/sumanth/Devops/MymavenWebApp01/src/main/webapp]
[INFO] Building war: /home/sumanth/Devops/MymavenWebApp01/target/MymavenWebApp01.war
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MymavenWebApp01 ---
[INFO] Installing /home/sumanth/Devops/MymavenWebApp01/target/MymavenWebApp01.war to /home/sumanth/.m2/repository/com/example/MymavenWebApp01/1.0-SNAPSHOT/MymavenWebApp01-1.0-SNAPSHO
T.war
[INFO] Installing /home/sumanth/Devops/MymavenWebApp01/pom.xml to /home/sumanth/.m2/repository/com/example/MymavenWebApp01/1.0-SNAPSHOT/MymavenWebApp01-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.192 s
[INFO] Finished at: 2026-04-15T23:57:23+05:30
```

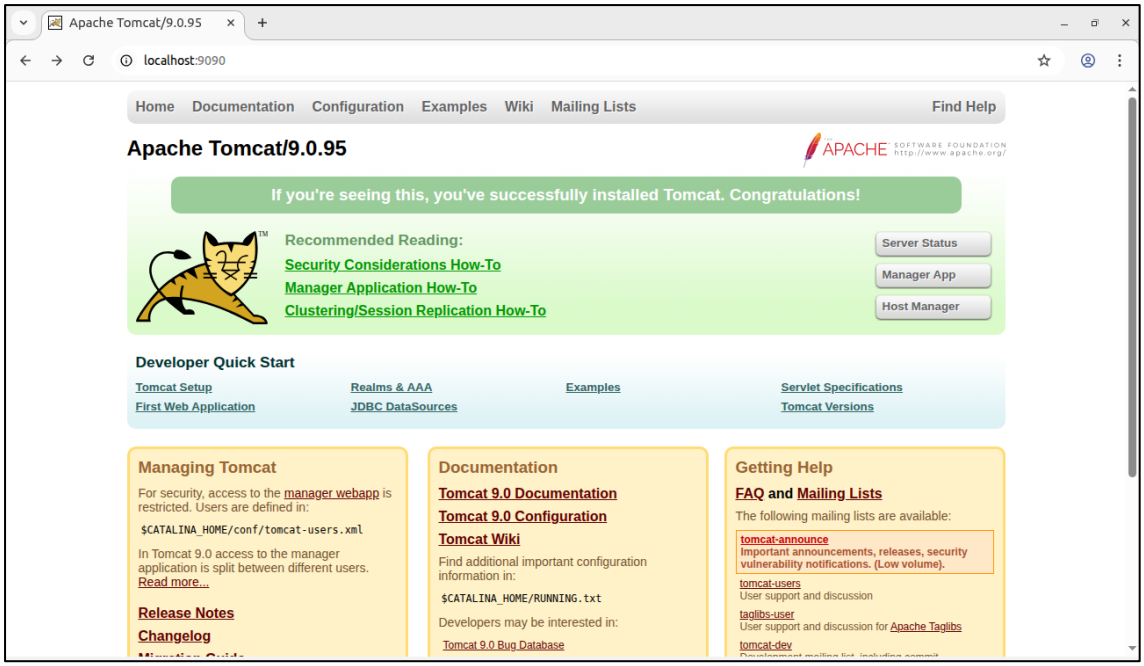
Step 4: Run **tree** in order to check if the War file exists

```
sumanth@sumanth-VirtualBox:~/Devops/MymavenWebApp01$ tree
.
├── Jenkinsfile
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── index.jsp
│           └── WEB-INF
│               └── web.xml
└── target
    ├── classes
    ├── maven-archiver
    │   └── pom.properties
    ├── MymavenWebApp01
    │   ├── index.jsp
    │   ├── META-INF
    │   └── WEB-INF
    │       ├── classes
    │       └── web.xml
    └── MymavenWebApp01.war

12 directories, 8 files
```

Step 5: **Run sudo /opt/tomcat/bin/startup.sh** in order to start the Tomcat Server

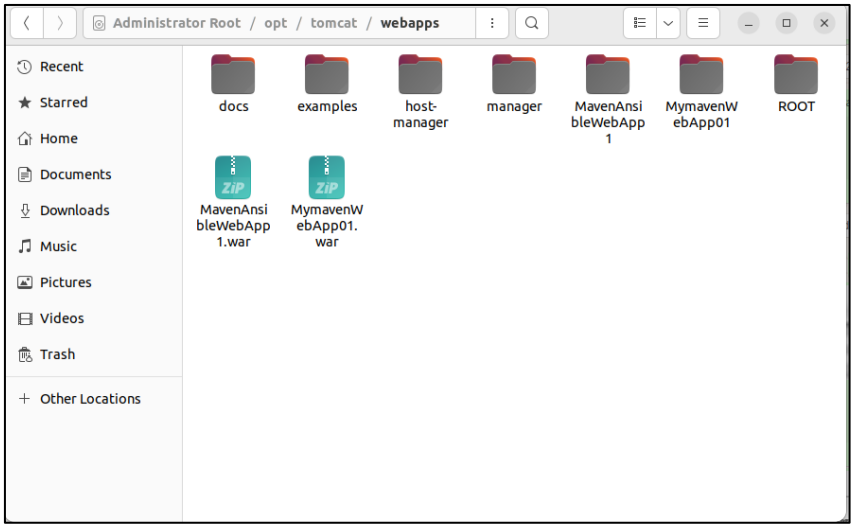
```
sumanth@sumanth-VirtualBox:~/Devops/MymavenWebApp01$ cd ..
sumanth@sumanth-VirtualBox:~/Devops$ cd ..
sumanth@sumanth-VirtualBox:~$ sudo /opt/tomcat/bin/startup.sh
[sudo] password for sumanth:
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
sumanth@sumanth-VirtualBox:~$
```

Step 6: Login in to the Respective Tomcat server with Username and Password for Manager App

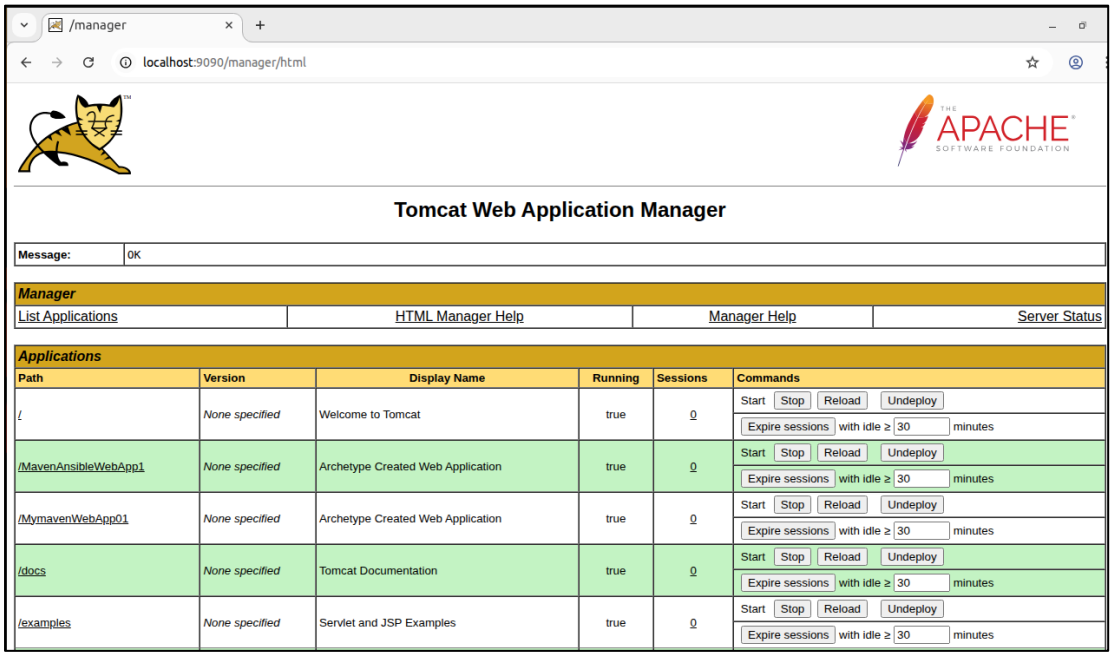


Step 7: Before Running the Web application on tomcat Server we need to change the **index.jsp** file also

Step 8: Now before running it, we need to copy the War file created and paste in the file path of tomcat file i.e. **/opt/tomcat/webapps.**After Pasting we can run the tomcat server easily

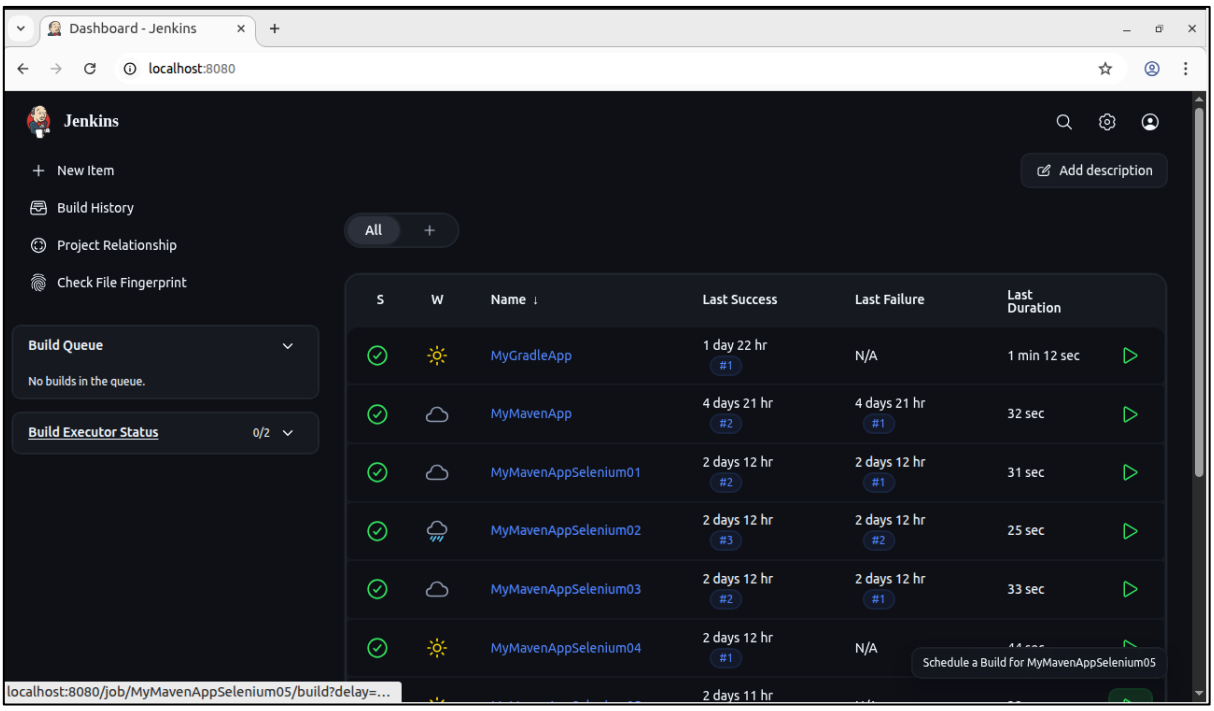


Step 9: Now we can run the tomcat server with the filename **MyMavenWebApp01** to get the output

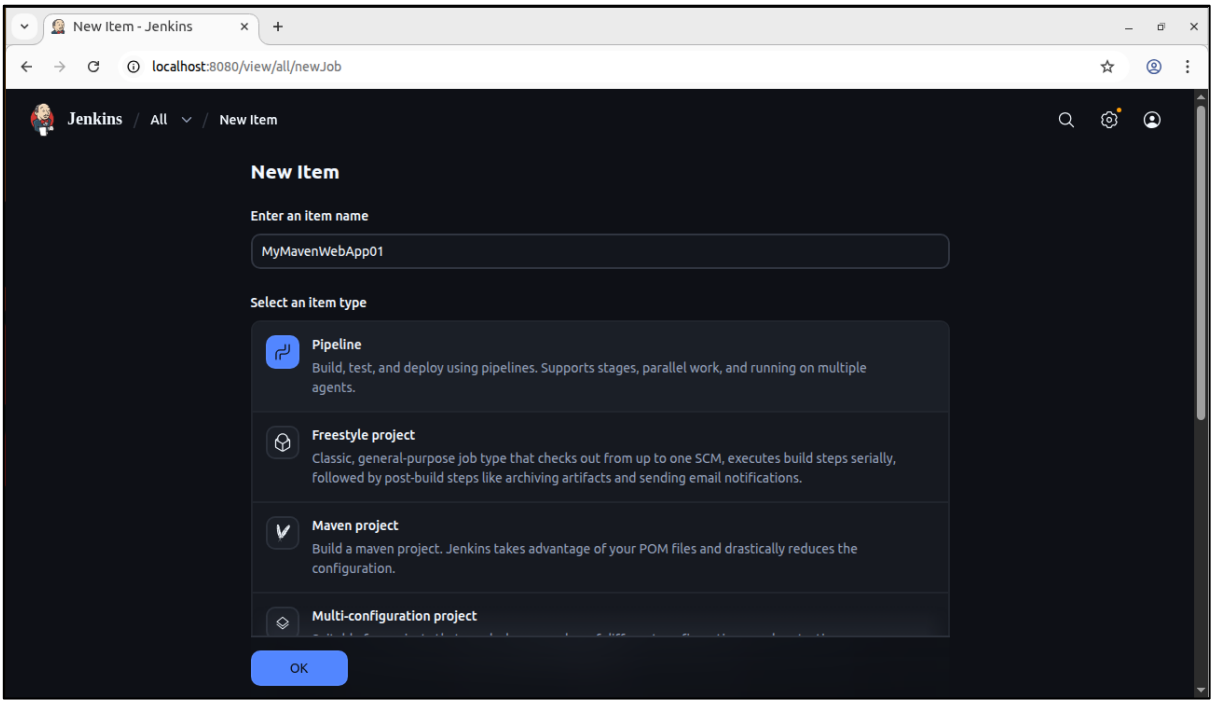


Jenkins Pipeline for MyMavenWebApp01

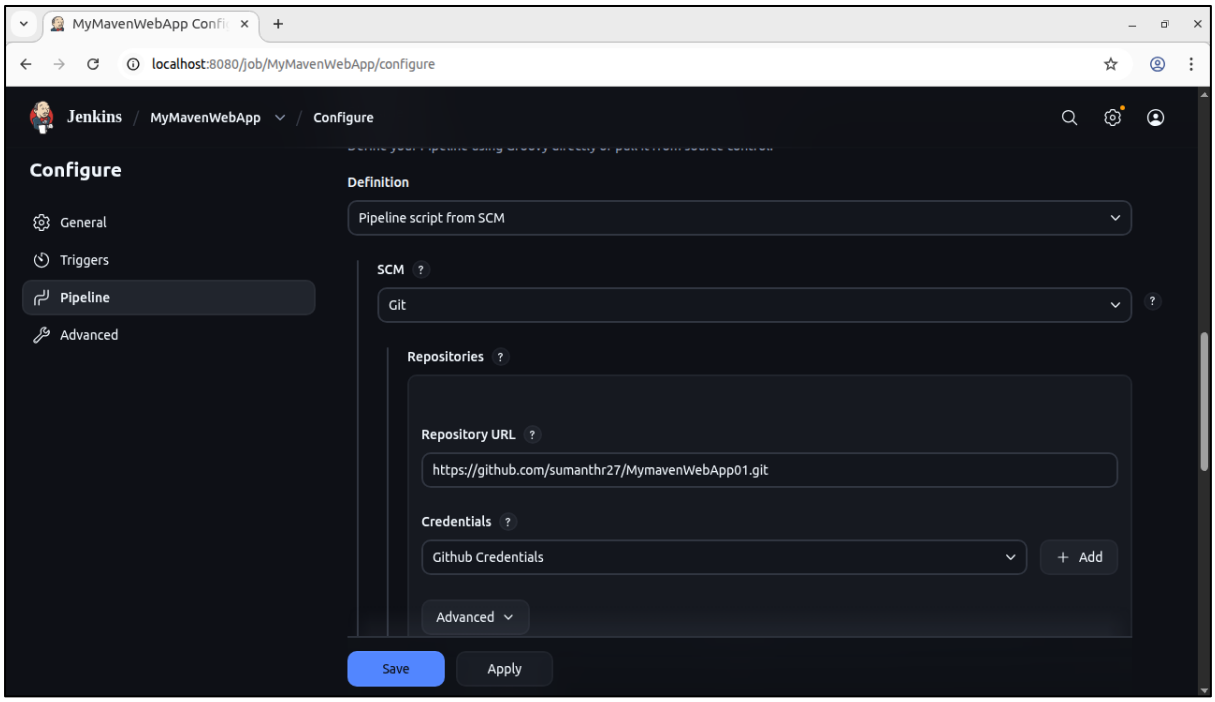
Step 1: Open the Jenkins Dashboard in order to create a new item



Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenWebApp



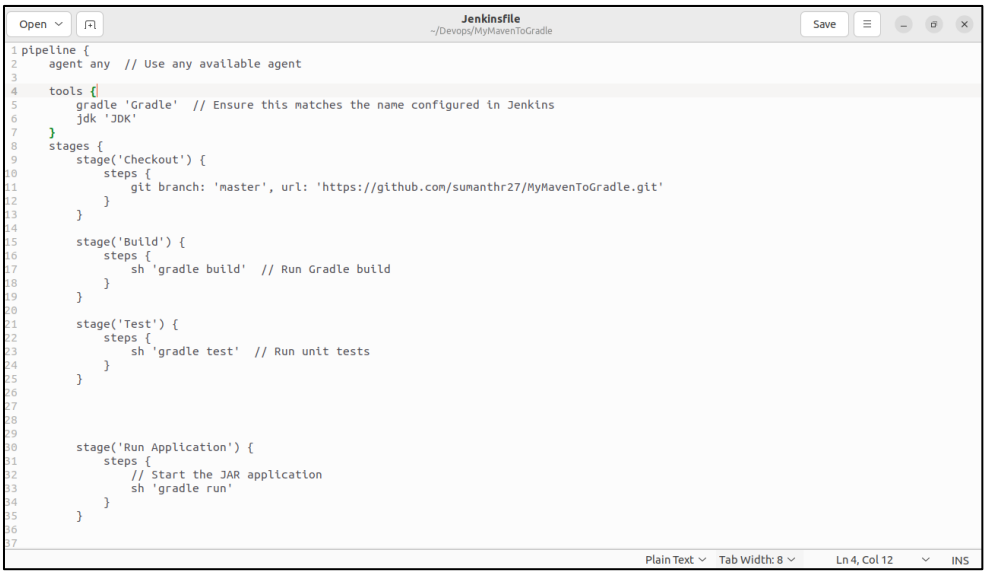
Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenWebApp01 and create an new file called “Jenkinsfile”

```
sumanth@sumanth-VirtualBox:~$ cd Devops
sumanth@sumanth-VirtualBox:~/Devops$ cd MymavenWebApp01
sumanth@sumanth-VirtualBox:~/Devops/MymavenWebApp01$ ls
Jenkinsfile  pom.xml  src  target
sumanth@sumanth-VirtualBox:~/Devops/MymavenWebApp01$ gedit Jenkinsfle
```

Step 5: Paste the available code inside the Jenkinsfile

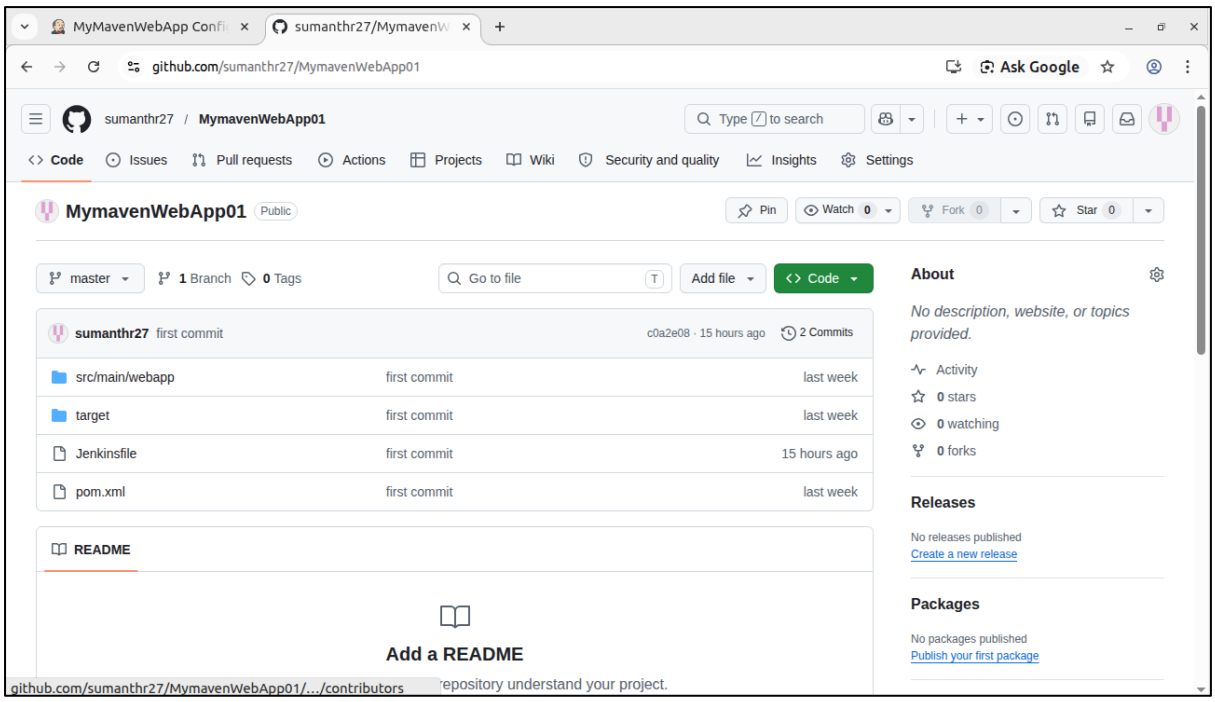



```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/sumanthr27/MyMavenToGradle.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }



        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
            }
        }
```

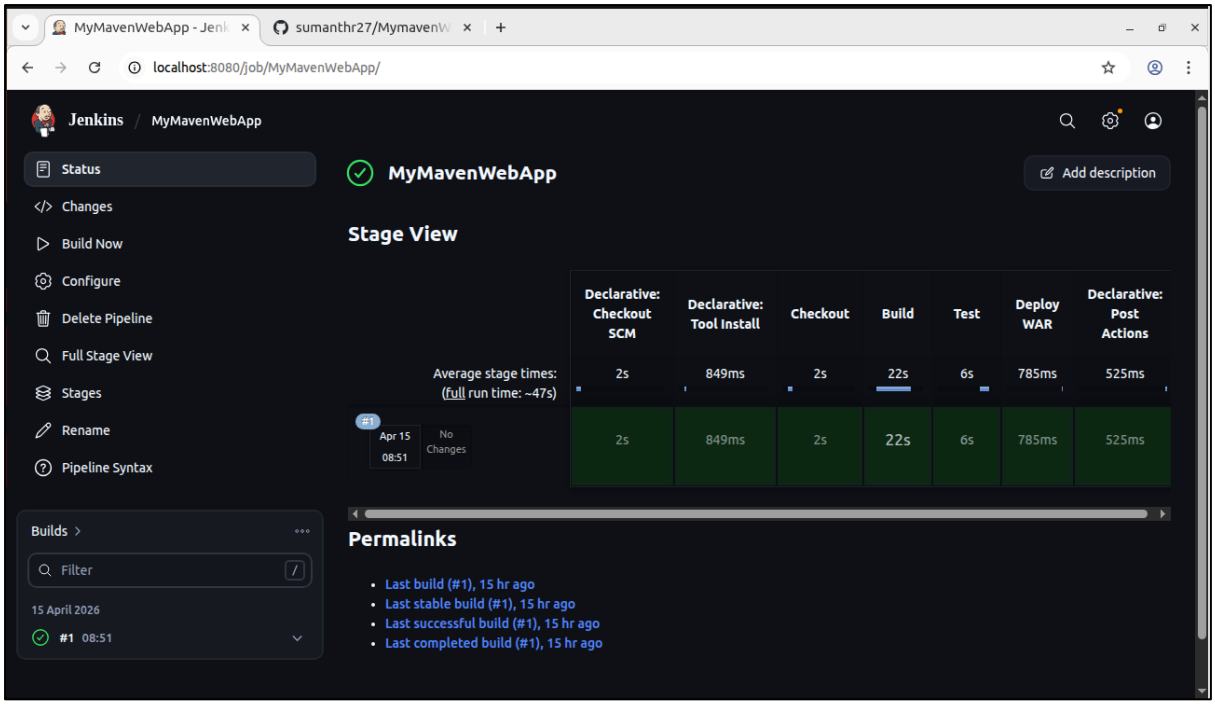
Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.


MyMavenWebApp - Jenk
sumanthr27/MymavenW
localhost:8080/job/MyMavenWebApp/
Jenkins / MyMavenWebApp
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Pipeline Syntax
Builds
Filter
15 April 2026
#1 08:51
MyMavenWebApp
Add description
Stage View
Declarative: Checkout SCM
Declarative: Tool Install
Checkout
Build
Test
Deploy WAR
Declarative: Post Actions
Average stage times:
(full run time: ~47s)
2s
849ms
2s
22s
6s
785ms
525ms
#1
Apr 15
08:51
No Changes
Permalinks
Last build (#1), 15 hr ago
Last stable build (#1), 15 hr ago
Last successful build (#1), 15 hr ago
Last completed build (#1), 15 hr ago